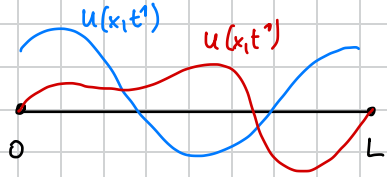
WAVE PROPAGATION IN 1D : $\Omega = (0,L)$

(1) $$\rho\, \partial^2_{tt} u(x,t) - \partial_x \left(\mu\, \partial_x u(x,t) \right) = f(x,t) \qquad x \in (0,L),\ t \in (0,T]$$

DENSITY $\rho(x) = \rho_0 > 0 \quad \forall x$

STIFFNESS $\mu(x) = \mu_0 > 0 \quad \forall x$



$$c(x) = \sqrt{\frac{\mu(x)}{\rho(x)}}$$ WAVE SPEED

- If $\rho(x) = \rho$, $\mu(x) = \mu \quad \forall x$ (ρ, μ CONSTANTS)

$$\Rightarrow \quad \partial^2_{tt} u - c^2 \partial_{xx} u = f/\rho \qquad \left[\partial_x u = \frac{\partial u}{\partial x} = u_x \right]$$ NOTATION

ACCELERATION

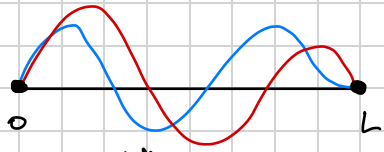
(1) $$\begin{cases} \rho\, \partial^2_{tt} u - \partial_x(\mu \partial_x u) = f & x \in (0,L)\ \ t \in (0,T] \\ + \text{ INITIAL CONDITIONS} \\ \partial_t u(x,0) = u_1(x) & \forall x \in (0,L) \\ u(x,0) = u_0(x) & \forall x \in (0,L) \\ + \text{ BOUNDARY CONDITIONS} \end{cases}$$

(BEHAVIOUR OF u OR u_x in $x = 0, L$)

1) DIRICHLET BC

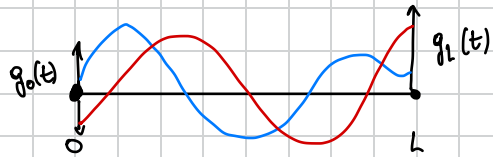
$$u(0,t) = g_0(t) \qquad u(L,t) = g_L(t)$$

- If $g_0(t) = g_L(t) \equiv 0$



THE STRING IS NOT ALLOWED TO MOVE IN 0 AND L

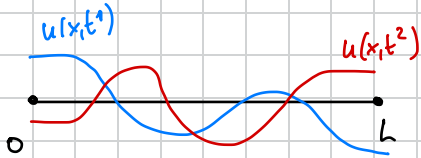
- Generally



2) NEUMANN CONDITIONS

$$\mu\, \partial_x u(0,t) = h_0(t) \qquad \mu\, \partial_x u(L,t) = h_L(t) \qquad \forall t \in [0,T]$$

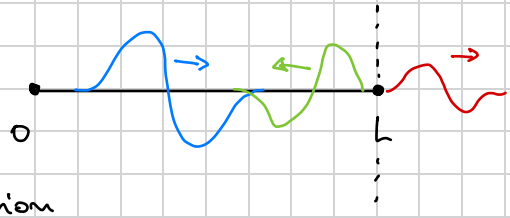
- $h_0(t) \equiv h_L(t) \equiv 0$



3) IMPEDANCE CONDITIONS

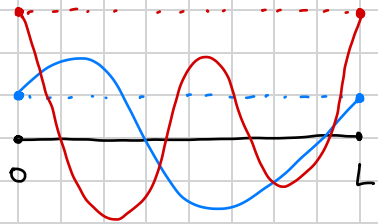
$$\begin{cases} \dfrac{1}{c}\, \partial_t u(0,t) - \alpha\, \partial_x u(0,t) = 0 \\[2ex] \dfrac{1}{c}\, \partial_t u(L,t) + \alpha\, \partial_x u(L,t) = 0 \end{cases}$$

- If $0 < \alpha < 1$ Reflection



- If $\alpha = 1$ Perfect transmission
  (ABSORBING OR TRANSPARENT B.C.)

4) PERIODIC BOUNDARIES

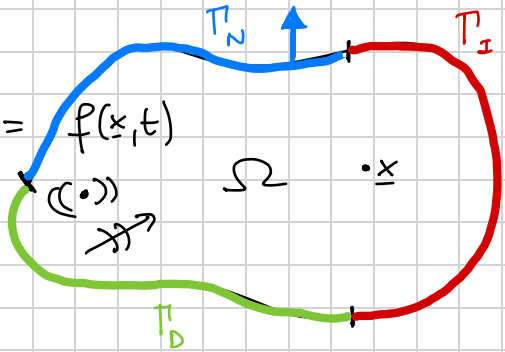


$$u(0,t) = u(L,t)$$

$$? \quad u_x(0,t) = -u_x(L,t)$$

(INFINITE STRING !)

IN MORE THAN 1D



$$\begin{cases} \rho\, \partial^2_{tt} u(\underline{x},t) - \nabla\cdot\left(\mu \nabla u(\underline{x},t)\right) = f(\underline{x},t) \\ \partial_t u(\underline{x},0) = u_1(\underline{x}) \quad \forall \underline{x} \in \Omega \\ u(\underline{x},0) = u_0(\underline{x}) \quad \forall \underline{x} \in \Omega \\ + \text{ BOUNDARY CONDITIONS} \end{cases}$$

$\partial\Omega$ = BOUNDARY OF Ω $\qquad \partial\Omega = \Gamma_D \cup \Gamma_N \cup \Gamma_I$

1) DIRICHLET CONDITIONS

$$u(\underline{x},t) = g(\underline{x},t) \qquad \forall \underline{x} \in \Gamma_D \qquad \forall t \in (0,T]$$

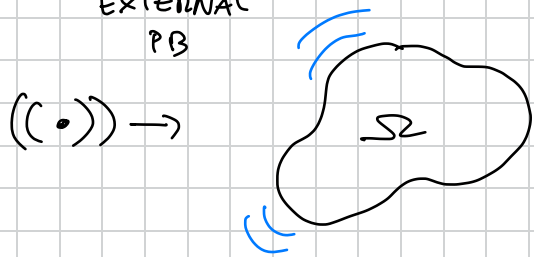
2) NEUMANN CONDITIONS

$$\mu \nabla u(\underline{x},t)\cdot \underline{n} = h(\underline{x},t) \quad \forall \underline{x} \in \Gamma_N \quad \forall t \in (0,T]$$

3) IMPEDANCE CONDITIONS

$$\frac{1}{c}\partial_t u(\underline{x},t) + \alpha \nabla \underline{u}\cdot \underline{n} = 0$$

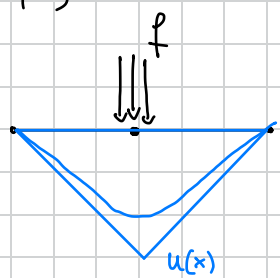
EXTERNAL PB



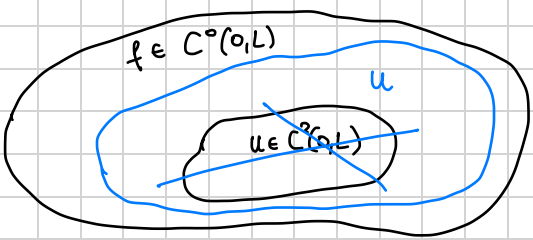
TO UNDERSTAND THE FINITE ELEMENT METHOD (FEM) WE CONSIDER THE FOLLOWING (STATIONARY) PROB.

STRONG FORMULATION

$$(P)\quad \begin{cases} \rho\, u(x) - \partial_x(\mu\, \partial_x u) = f & \text{in } (0,L) \\ \mu\, \partial_x u(0) = \mu\, \partial_x u(L) = 0 \\ \mu u'(0) = \mu u'(L) = 0 \end{cases}$$



$$f \in C^0(0,L) \implies u \in C^2(0,L)$$



$$\frac{\partial^2 u}{\partial x^2}$$

WEAK FORMULATION. Take $v \in C^1(0,L)$, multiply (P1) for v and integrate by parts

$$\int_0^L \rho u v \, dx - \underbrace{\int_0^L \partial_x(\mu \partial_x u) v \, dx}_{\text{INTEGRATE BY PARTS}} = \int_0^L f v \, dx$$

$$\partial_x(\mu \partial_x u) = (\mu u'(x))' \qquad \int_a^b f' g \, dx = [f g]_a^b - \int_a^b f g' \, dx$$

$$\Rightarrow -\int_0^L (\mu u')' v \, dx = -[\mu u' v]_0^L + \int_0^L \mu u' v' \, dx$$

$$= \underbrace{-\mu u'(L) \cancelto{0}{v(L)} + \mu u'(0) \cancelto{0}{v(0)}}_{\text{BOND. TERM}} + \int_0^L \mu u' v' \, dx$$

Find $u \in C^1(0,L)$ s.t.

$$\underbrace{\int_0^L \rho u v \, dx}_{\substack{\text{A} \\ \in \mathbb{R} < +\infty}} + \underbrace{\int_0^L \mu u' v' \, dx}_{\substack{\text{B} \\ < +\infty}} = \underbrace{\int_0^L f v \, dx}_{\substack{\text{C} \\ < +\infty}} \quad \forall v \in C^1(0,L)$$

If $C^1(0,L)$ the correct space?

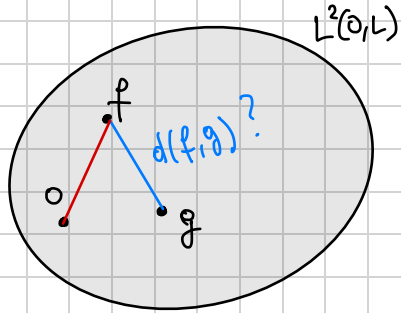
For which A, B & C are $< +\infty$?

LEBESGUE SPACES $L^p(\Omega)$ $(p=2)$

$$L^2(0,L) = \left\{ v : (0,L) \to \mathbb{R} \text{ s.t. } \int_0^L f^2 \, dx < +\infty \right\}$$

SQUARED INTEGRABLE FUNCTIONS

$$(f,g) = \int_0^L f g \, dx \qquad \text{SCALAR PRODUCT}$$

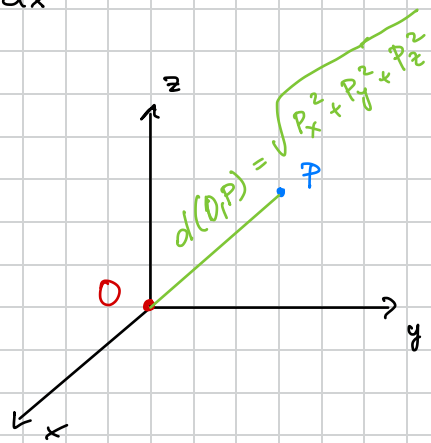


$$(f,f) = \int_0^L f \cdot f \, dx = \int_0^L f^2 \, dx$$

$$\| f \|^2_{L^2(0,L)} := (f,f) = \int_0^L f^2 \, dx$$

$$\| f \|^2_{L^2(0,L)} = d(0,f)$$

$$d(f,g) = \| f - g \|_{L^2(0,L)}$$



CAUCHY-SCHWAR INEQUALITY

If $f, g \in L^2(0,L)$

$$\Rightarrow \int_0^L \underbrace{|f \cdot g|}_{< +\infty} \leq \overset{< +\infty}{\| f \|_{L^2(0,L)}} \overset{< +\infty}{\| g \|_{L^2(0,L)}}$$

$$\int_0^L \rho \overset{< +\infty}{u v} \, dx + \overbrace{\int_0^L \mu \, \partial_x u \, \partial_x v \, dx}^{< +\infty} = \overbrace{\int_0^L f v \, dx}^{< +\infty}$$

$u, v \in L^2(0,L)$ $\qquad$ $u', v' \in L^2(0,L)$ $\qquad$ $f, v \in L^2(0,L)$

$$\int_0^L \rho u v \, dx \leq \int_0^L |\rho u v| \, dx \leq \rho_{MAX} \int_0^L |u v| \, dx \leq \rho_{MAX} \underset{< +\infty}{\| u \|} \underset{< +\infty}{\| v \|}$$

$u, v, u', v', f \in L^2(0,L)$

$$H^{1}(0,L) = \{ v : (0,L) \to \mathbb{R} \text{ s.t. } v \,\&\, v^{1} \in L^2(0,L) \}$$

HILBERT SPACE

$$\|v\|^2_{H^1(0,L)} = \|v\|^2_{L^2(0,L)} + \|v'\|^2_{L^2(0,L)}$$

WEAK FORMULATION Given $f \in L^2(0,L)$

Find $u \in H^1(0,L)$ s.t.

$$\int_0^L \rho u v \, dx + \int_0^L \mu u' v' \, dx = \int_0^L f v \, dx \quad \forall v \in H^1(0,L)$$

$H^1(0,L) \equiv C^0(0,L)$ in 1D

I can rewrite as:

(WP)

Find $u \in H^1(0,L)$ s.t. $a(u,v) = F(v) \quad \forall v \in H^1(0,L)$

where $a(u,v) = \int_0^L \rho u v \, dx + \int_0^L \mu u' v' \, dx$

$$F(v) = \int_0^L f v \, dx$$

$$a : \underset{u}{H^1(0,L)} \times \underset{v}{H^1(0,L)} \longrightarrow \mathbb{R}$$ BILINEAR FORM

$$F : H^1(0,L) \longrightarrow \mathbb{R}$$ LINEAR FUNCTIONAL

LAX-MILGRAM LEMMA:

Suppose that V is a Hilbert space $(V = H^1(0,L))$ and suppose that $a : V \times V \to \mathbb{R}$ is a bilinear form, $F : V \to \mathbb{R}$ is a linear functional, and:

1) $\exists\, \alpha > 0$ s.t. $a(u,u) \geq \alpha \|u\|_V$ COERCIVITY
2) $\exists\, M > 0$ s.t. $|a(u,v)| \leq M \|u\|_V \|v\|_V$ CONTINUOUS
3) $\exists\, C > 0$ s.t. $|F(v)| \leq C \|v\|_V$ CONTINUOUS

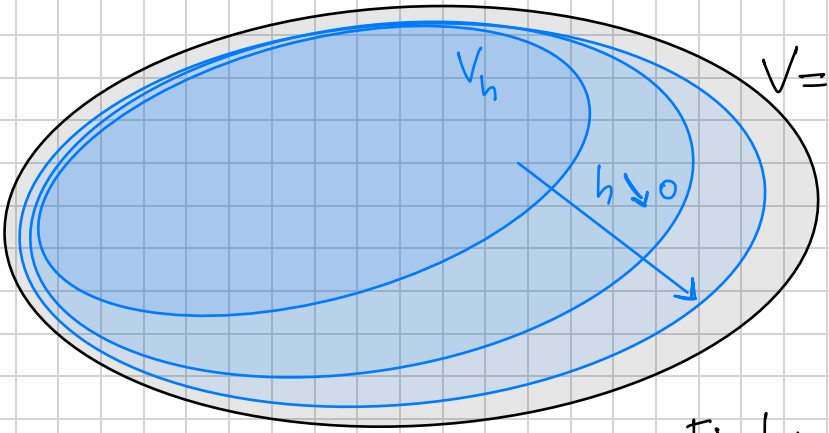
$\Rightarrow$ $\exists!\, u \in V$ to $a(u,v) = F(v)$ $\forall v \in V$

and $$\|u\|_V \leq \frac{1}{\alpha} \sup_{v \neq 0} |F(v)|$$

NOTE: It is possible to prove that (WP) is well-posed through the Lax-Milgram Lemma!

DETAILS if required.

FINITE ELEMENT METHOD



$V = H^1(0,L)$

$\dim V = +\infty$

Find $u \in V$ s.t.

$a(u,v) = F(v) \quad \forall v \in V$

- $V_h \subset V$
- $\dim V_h = N_h < +\infty$
- $\lim_{h \to 0} V_h = V$

IDEA: Reformulate (WF) in V_h

$$\boxed{\text{Find } u_h \in V_h \subset V \text{ s.t. } a(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h}$$

FINITE ELEMENT FORMULATION

$\Rightarrow \quad \exists!\, u_h \in V_h$ solution

LM LEMMA

HOW CAN I FIND $u_h \in V_h$?

- Suppose that $\exists$ a basis for V_h and call it $\{\varphi_j(x)\}_{j=1}^{N_h}$

$\Rightarrow$ Any $u_h \in V_h$ can be written

$$u_h(x) = \sum_{j=1}^{N_h} u_j \varphi_j(x) \qquad u_j \in \mathbb{R}$$

$$a(u_h, v_h) = F(v_h)$$

$$\Updownarrow$$

$$a\left(\sum_{j=1}^{N_h} u_j \varphi_j(x), v_h(x)\right) = F(v_h(x)) \quad \forall v_h$$

$$\Updownarrow \quad v_h = \varphi_i \quad \forall i$$

$$a\left(\sum_{j=1}^{N_h} u_j \varphi_j(x), \varphi_i(x)\right) = F(\varphi_i(x)) \quad \forall i = 1, \dots, N_h$$

$$\Updownarrow$$

$$\sum_{j=1}^{N_h} u_j \underbrace{a(\varphi_j(x), \varphi_i(x))}_{1} = \underbrace{F(\varphi_i(x))}_{\in \mathbb{R}} \quad \forall i = 1, \dots, N_h$$

$$= b_i$$

$$a_{ij} = \int_0^L p\,\varphi_j \varphi_i + \int_0^L \mu\,\varphi_j' \varphi_i' \in \mathbb{R}$$

$$\Updownarrow$$

$$\sum_{j=1}^{N_h} a_{ij} u_j = b_i \qquad \forall i = 1, \dots, N_h$$

LINEAR SYSTEM

$$\begin{cases} a_{11} u_1 + a_{12} u_2 + \dots + a_{1N_h} u_{N_h} = b_1 \\ a_{21} u_1 + a_{22} u_2 + \dots + a_{2N_h} u_{N_h} = b_2 \\ \vdots \\ a_{N_h 1} u_1 + a_{N_h 2} u_2 + \dots + a_{N_h N_h} u_{N_h} = b_{N_h} \end{cases}$$

N_h EQs $\qquad$ N_h UNKNOWNS

$a_{ij} \longrightarrow A \in \mathbb{R}^{N_h \times N_h}$

$b_i \longrightarrow \underline{b} \in \mathbb{R}^{N_h}$

$$\Updownarrow$$

Find $\underline{u} \in \mathbb{R}^{N_h}$ s.t. $A\underline{u} = \underline{b}$

$$\underline{u} = A^{-1}\underline{b} \iff \det(A) \neq 0$$

A is non singular because $a(\cdot,\cdot)$ is COERCIVE !

$a(\cdot,\cdot)$ COERCIVE $\Longrightarrow$ A is POSITIVE DEFINITE

$$\forall \underline{x} \in \mathbb{R}^n \quad \underline{x}^T A \underline{x} \geq 0$$

$$\underline{x}^T A \underline{x} = 0 \iff \underline{x} = \underline{0}$$

$$u_h \in V_h \longleftrightarrow u_J \in \mathbb{R}^{N_h}$$

$$\underline{u}^T A \underline{u} = \sum_{i,j=1}^{N_h} u_i A_{ij} u_j = \sum_{i,j}^{N_h} u_i \, a(\varphi_j, \varphi_i) u_j$$

$$= a\left(\sum_i u_i \varphi_i, \sum_j u_j \varphi_j\right)$$

$$= a(u_h, u_h) \geq \alpha \|u_h\|^2_{L^2(0,L)} = \alpha \underline{u}^2 \geq 0$$

$= 0$ only if $\underline{u} = \underline{0}$

WEAK FORM $\longrightarrow$ FE FORM. $\longleftrightarrow$ ALGEBRAIC SYSTEM

$u \in V$ $\qquad$ $u_h \in V_h \subset V$ $\qquad$ $A\underline{u} = \underline{b}$

$d(u, u_h)$? WHAT IS THE APPROXIMATION ERROR ?

$$\begin{cases} a(u, v_h) = F(v_h) & \forall v_h \in V_h \\ a(u_h, v_h) = F(v_h) & \forall v_h \in V_h \end{cases}$$

$$\boxed{a(u-u_h, v_h) = 0 \qquad \forall v_h \in V_h}$$

CEÁ LEMMA

COERCIVITY

$$\alpha \|u-u_h\|^{\cancel{2}}_{H^1(0,L)} \leq a(u-u_h, u-u_h)$$

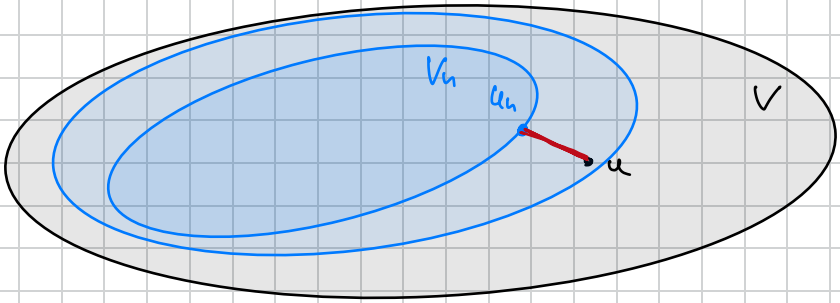
$$= a(u-u_h, u-u_h \pm v_h)$$

$$= a(u-u_h, u-v_h) + \cancel{a(u-u_h, \underbrace{v_h-u_h}_{\in V_h})}^{\,0}$$

$$\leq M \cancel{\|u-u_h\|_{H^1(0,L)}} \|u-v_h\|_{H^1(0,L)}$$

$$\boxed{\|u-u_h\|_{H^1(0,L)} \leq \inf_{v_h} \frac{M}{\alpha} \|u-v_h\|_{H^1(0,L)} \qquad \forall v_h}$$

BEST APPROXIMATION OF $u \in V_h$

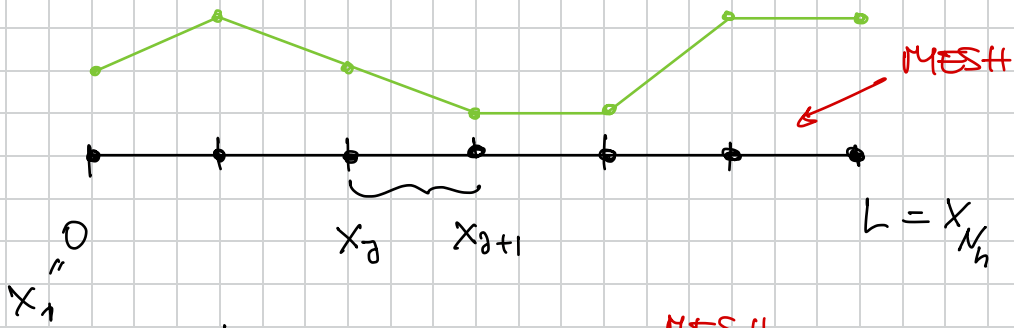


$$\lim_{h\to 0} \| u - u_h \|_{H^1(0,L)} = \frac{M}{\alpha} \lim_{h\to 0} \| u - v_h \|_{H^1(0,L)} = 0$$

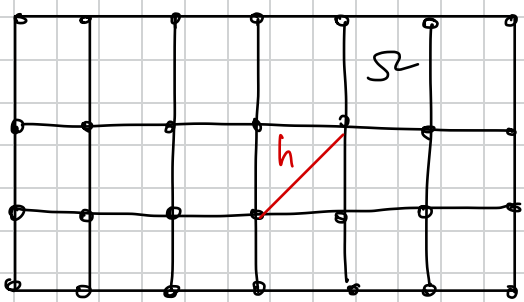
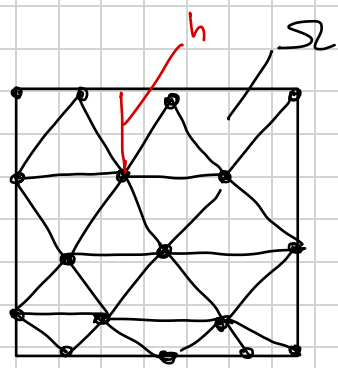
$$V_h \longrightarrow V$$

What is "h" ?

1D)



$$h = x_{j+1} - x_j \longrightarrow \text{MESH SIZE}$$

V_h = piece-wise linear function $\subset H^1(0,L)$

2D)